05 · Case 3, part 3: her four lung micrographs

The words of this track, as it says them. Bold brown is the examiner.

Exam case 3, continued. The four lung micrographs. Ten marks.

Her four lung micrographs, A to D, from the case study workbook. Name the part of the airway each one shows. Give its lining, and say whether cartilage and smooth muscle are present.

This table isn't starred. But her exam case study helpline links a deck called Case Study Lung Histology for Exam under this case, and she says the booklet's diagrams will be in the exam. So learn all four. The wording of the question and the letters here follow the workbook's order.

The hook: cartilage stops at the bronchiole. Muscle stops at the alveoli.

Letter A, the trachea. It's lined with ciliated pseudostratified columnar epithelium. Cartilage: present. Smooth muscle: present.

Letter B, a bronchus. Also ciliated pseudostratified columnar epithelium. Smooth muscle: present. Cartilage: present.

Letter C, a bronchiole. Lined with cuboidal epithelium. Cartilage: absent. Smooth muscle: present. Smooth muscle with no cartilage to hold the tube open is what lets a bronchiole squeeze down in asthma.

Letter D, the alveoli. Mostly type one pneumocytes, which are squamous epithelium: thin, for gas exchange. A smaller number of type two pneumocytes, which make surfactant. And macrophages, which keep the alveoli clean. No cartilage, and no muscle.

And her last line ties it back to Ane. The smooth muscle is relaxed by bronchodilators: the reliever. Cortisol, taken daily, is the preventer, and it reduces the inflammation.

Once more. The four micrographs.

The trachea and a bronchus: ciliated pseudostratified columnar epithelium, with cartilage and smooth muscle. A bronchiole: cuboidal epithelium, with smooth muscle but no cartilage. The alveoli: type one pneumocytes, squamous epithelium for gas exchange; type two pneumocytes for surfactant; macrophages to keep them clean; and no cartilage or muscle. Bronchodilators relax the smooth muscle; daily cortisol, the preventer, reduces the inflammation.